Summary Report: Candidate literary paper to be used

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1 Paper 1: Optimum government size and economic growth in case of Indian states: Evidence from panel threshold model

1.1 Summary

This paper investigates the relationship between optimum government size and economic growth using data of Indian states during 1990-91 to 2017–18. Our results derived from panel threshold regression model show a positive and significant impact of government size on economic growth within the estimated thresholds for both aggregate and sub-panels based on income and regions. Once the government size moves above the upper threshold level, then its impact declines and turns to be insignificant. Thus, our findings suggest the policymakers for maintaining the government size within the thresholds limit. Akram and Rath 2020

1.2 Metadata

1. **Link:** https://doi.org/10.1016/j.econmod.2019.09.015
2. **Journal Name:** Economic Modelling
3. **Rank:** Q1

1.3 Data sources

- Indian states during 1990-91 to 2017–18
- Gross State Domestic Products (GSDP, 2011–12) and expenditures (such total expenditure, social sector expenditure
- revenue expenditure, and capital expenditure) data collected from Economic and Political Weekly Research Foundation (EPWRF).

1.4 Methodology

theoretical model which is proposed by Barro (1990, 1991).

$$Y(t) = AK(t)^{\beta} G(t)^{1-\beta}, \quad 0 < \alpha < 1$$

The thresholds model is as follows:

$$\delta y_{i,t} = \alpha_{)} + \pi_i t + \beta_i y_{t-1} + \sum_{j=1}^{k} \psi_{i,j} \delta y_{i,t-1} + \epsilon_{i,t}$$

1.5 Conclusions

- **Objective:** The study addresses the literature gap regarding sub-national economic growth by examining the relationship between optimum government size and growth across major Indian states (1990–91 to 2017–18).
- **Methodology:** A panel threshold regression model was employed, identifying two significant thresholds for the aggregate panel and most sub-panels.
- **Optimal Thresholds (Aggregate):** Government size has a positive and significant effect on economic growth specifically when it lies between the lower and upper thresholds of 4.85% and 5.09%.
- **Category Variations:**
  - **NSC States:** Estimated optimal threshold range is 4.55%–6.07%.
  - **SC States:** Estimated optimal threshold range is significantly higher at 34.70%–38.55%.
- **Diminishing Returns:** Once government size exceeds the upper threshold, the coefficient of growth declines and becomes statistically insignificant.

- **Expenditure Composition:** Findings remain consistent when disaggregated into revenue, capital, and social sector expenditures, showing positive growth associations within threshold limits.
- **Policy Recommendations:** State governments should utilize fiscal rules and budgetary discipline to maintain an ideal size, ensuring that Keynesian deficit spending remains within the "optimum" range to avoid unfavorable growth consequences.

2 Paper 2: Does too much government spending depress the economic development of transition economies? Evidences from dynamic panel threshold analysis

2.1 Summary

This paper investigates the relationship between government size and economic growth and determines the optimal level of government spending to maximize economic growth. The paper applies a dynamic panel data analysis based upon a threshold model to test the threshold effect of government spending in 26 transition economies over the period spanning 1993–2016. According to the analysis results, government expenditures have a threshold effect on economic growth, and there is a non-linear relationship depicted as an Armey curve in these transition economies. The findings indicate that a government size above the threshold government spending level adversely affects economic growth, while a government size below the threshold level has a positive effect. Furthermore, there is a statistically significant relationship between the two variables above and below that optimal level, even if we divide the sample into developed and developing countries. Our findings suggest that governments in transition economies should consider optimal government size at around the estimated threshold level to support sustainable economic growth.Aydin and Esen 2019

2.2 Metadata

1. **Link:** https://doi.org/10.1080/00036846.2018.1528335
2. **Journal Name:** Applied Economics
3. **Rank:** Q2

2.3 Methodology

$$Y_{it} = \alpha_0 + \alpha_i initial_{it} + \alpha_2 Gov_{it} + \beta x_{it} + \epsilon_{it}$$

Where:

- Y_{it}: Real GDP per capita growth rate in country i at time t.
- α_0: Constant term.
- $initial_{it}$: Initial level of income in country i at time t.
- α_1: Coefficient measuring the effect of initial income on economic growth.
- Gov_{it}: Government spending level in country i at time t.
- α_2: Coefficient capturing the impact of government spending on economic growth.
- x_{it}: Vector of other macroeconomic control variables affecting economic growth.
- β: Vector of coefficients associated with the control variables.
- ε_{it}: White noise error term.

2.4 Conclusions

- The relationship between government expenditures and economic growth is widely debated, but has rarely been studied in the context of transition economies moving from centrally planned to free market systems.
- This study empirically examines threshold effects of government size on economic growth in 25 transition economies from 1993 to 2016.
- Using a threshold autoregressive (TAR) approach, the study avoids assuming a specific functional form and identifies optimal government size more accurately.
- Results show a non-linear relationship: government expenditures positively affect growth below a threshold, but this effect weakens and becomes negative (though insignificant) beyond it.
- The estimated threshold levels are 11.67 % of GDP for developing economies and 17.54 % for developed economies.
- Government intervention supports growth through education, health, inequality reduction, and institutional development, but excessive intervention leads to diminishing returns.
- Beyond the optimal size, government expansion may hinder growth due to taxation burdens, inefficiencies, corruption, rent-seeking, and crowding out of private investment.
- Transition economies require some degree of government intervention (e.g., liberalization, privatization, legal reforms) to support market functionality during the transition process.
- Sustainable growth depends on effective and limited government intervention; exceeding the optimal size does more harm than good.
- The study also highlights that government impact extends beyond expenditure size to include regulatory, legal, and bureaucratic interventions.

3 Paper 3: GOVERNMENT SIZE AND ECONOMIC GROWTH IN TURKEY: A THRESHOLD REGRESSION ANALYSIS

3.1 Summary

We examine the relationship between the government size and economic growth by using threshold regression model and quarterly data over the period 1998:1–2015:1 for Turkey. Our results provide a strong evidence for the existence of a non-linear relationship. The estimated threshold levels, as a percentage of GDP, are 16.5 for the government total expenditures, 12.6 for consumption expenditures and 3.9 for investment expenditures. We find that an increase in the government size leads to a significant rise (decline) in economic growth rate when the government size is below (above) the threshold level, confirming the predictions of Armey curve. Our findings have a clear policy implication: since the realized government consumption and total expenditures are well above the estimated threshold levels, a reduction in the government size would boost the growth rate. Iyidogan and Turan 2017

3.2 Metadata

1. **Link:** https://doi.org/10.18267/j.pep.600
2. **Journal Name:** Prague Economic Papers
3. **Rank:** Q3

3.3 Methodology

$$y_i = \alpha_0 + alpah_1(\frac{I_t}{Y_t}) + \alpha_2 Gov_t + u_t$$

- Y: GDP growth rate.
- C/Y: Government consumption expenditure (% of GDP).

- GI/Y: Government sector gross fixed capital formation (% of GDP).
- G/Y: Total government expenditure (% of GDP).
- GOV: The product of government expenditure growth and government size (government expenditure / GDP).
- I/Y: Private sector gross fixed capital formation (% of GDP).

3.4 Conclusions

- **Non-linear Relationship:** The study identifies a non-linear, non-monotonic relationship between government size and economic growth in Turkey (1998:1–2015:1), confirming the validity of the *Armey curve*.
- **Threshold Estimates:** Threshold levels as a share of GDP are estimated at 16.5% for total expenditures (G/Y), 12.6% for consumption expenditures (C/Y), and 3.9% for investment expenditures (GI/Y).
- **Growth Dynamics:** Government expansion significantly boosts economic growth when below the threshold level, but has a detrimental effect once the size exceeds these optimal limits.
- **Current Fiscal Status:** Realized total and consumption expenditures in Turkey have consistently remained above the estimated thresholds, while investment expenditures exceeded the threshold following the 2003–2009 stabilization period.
- **Policy Implications:** Findings suggest that to boost growth, policy makers should prioritize spending efficiency and restrain unproductive expenditures, particularly given future fiscal pressures such as an ageing population.
- **Future Fiscal Risk:** Projected increases in health and social security spending due to demographic shifts may further expand government size beyond optimal levels if structural reforms are not implemented.

4 Paper 4: THE EFFECT OF GOVERNMENT SPENDING ON ECONOMIC GROWTH

4.1 Summary

Theoretical models suggest a non-linear relationship between government size and long-run economic growth. However, testing this hypothesis empirically in cross-country studies is complicated by the endogeneity of government spending and the accurate identification of inflexion points. This paper examines the non-linear hypothesis by incorporating threshold analysis in a cross-country growth regression. The methodology utilizes a sample-splitting framework and follows an objective strategy for identifying and testing changes in the slope. The results provide evidence in support of the non-linear hypothesis for a broad panel of countries. Christie 2014

4.2 Metadata

1. **Link:** https://doi.org/10.1111/j.1467-8586.2012.00438.x
2. **Journal Name:** Bulletin of Economic Research
3. **Rank:** Q3

4.3 Methodology

$$Growth_{it} = \beta_0 X_{it} + \beta_1 Gov^{\star}_{it} I(Gov_{it} \leq \lambda) + \beta_1 Gov^{\star}_{it} I(Gov_{it} > \lambda)$$
$$u_{it} = \mu_i + \theta_t + \epsilon_{it}$$

- $Growth_{it}$: GDP growth rate for country i at time t.
- X_{it}: Vector of control variables (such as I/Y).
- μ_i: Country-specific fixed effect.

- θ_t: Time fixed effect.
- ε_{it}: Normally distributed error term.
- $I(\cdot)$: Indicator function; takes the value of one when the condition inside parentheses is satisfied.
- λ: The threshold value to be determined within the model.
- $S(\lambda) = \hat{u}(\lambda)'\hat{u}(\lambda)$: Residual sum of squares of the model estimated for a threshold level λ.
- $\hat{\lambda} = \arg\min_{\lambda} S(\lambda)$: The optimal threshold value that minimizes the residual sum of squares.

4.4 Conclusions

- **Non-linear Hypothesis:** The study confirms the *Barro non-linear hypothesis* using Hansen (2000) threshold regression on a panel of 136 countries, finding that the impact of government size on growth is contingent on its total share of GDP.
- **Threshold Estimates:** A universal threshold is identified at 33 % of GDP for the full sample; however, this optimal point is lower for developed countries, estimated between 26 % and 32 %.
- **Growth Dynamics:** Government expenditure below the threshold typically yields positive growth effects particularly when focusing on productive elements whereas spending exceeding 33 % of GDP exerts a strong negative effect on long-run growth.
- **Institutional Mitigation:** The quality of governance acts as a critical moderator; high-quality government services can dampen the negative effects of a large public sector, making non-linearities more pronounced in countries with low institutional effectiveness.
- **Developmental Variation:** While many developed nations currently exceed optimal size (over 75 % of the sample), many developing countries in Asia and the ECA remain below the threshold and could benefit from strategic budget increases.
- **Policy Implications:** Policy makers are advised to create ”fiscal space” for productive capital in developing regions while exercising restraint and prioritizing structural efficiency in developed economies to avoid the detrimental effects of over-expansion.

5 Paper 5: Optimal government size and economic growth in France (1871-2008)

5.1 Summary

This paper analyses the effect of public expenditure on economic growth from both a theoretical and an empirical point of view. Given that the economic literature supplies numerous and conflicting views on the topic, the article offers a framework combining both theories of market failures and State failures to account for an inverted U-shaped relation between government size and GDP growth. The empirical contribution is to provide evidence through a long time-series analysis of the existence of such a relation on the period 1871-2008 for France, which offers one of the longest stable democratic periods to analyse. Facchini and Melki 2011

5.2 Metadata

1. **Link:** https://shs.hal.science/halshs-00654363v1
2. **Journal Name:** N/A
3. **Rank:** N/

5.3 Methodology

$$\log(GDP\_FR_t) = C + \alpha_1 \log(GSIZE_t) + \alpha_2 \log(GSIZE_t)^2 + \alpha_3 \log(GDP\_EU_t) \\ + \alpha_4 \log(OPEN_t) + \alpha_5 \log(EMPLOY_t) + \alpha_6 \log(TAX_t) + \alpha \log(POP_t) + \epsilon_t$$

- GDP_FR_t: The annual GDP of France at time t.
- $GSIZE_t$: The size of government, expressed as the share of public expenditure (central state, social protection, and local public authorities) in total GDP at time t.
- GDP_EU_t: The annual GDP of 11 European trade partners at time t.
- $OPEN_t$: The degree of openness of the economy, defined as the sum of exports and imports as a percentage of total GDP at time t.
- $EMPLOY_t$: The national level of employment, measured by the average number of workers per year in France at time t.
- TAX_t: The total tax rate, measured at the national level at time t.
- POP_t: The total population of France at time t.
- ϵ_t: The error term at time t.
- C: The constant term in the regression model.
- $\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5, \alpha_6, \alpha_7$: The coefficients of the explanatory variables in the quadratic model.
- $\log(\cdot)$: Natural logarithm transformation applied to the respective variables.

5.4 Conclusions

- The study investigates theoretically and empirically the relationship between government size and economic growth, proposing a non-linear, inverted U-shaped relationship.
- The theoretical contribution combines State failure and market failure theories into a unified framework, decomposing the Armey curve into:
  - Costs arising from State failures.
  - Benefits derived from correcting market failures.
- Empirical analysis using long-term time-series data for France (1871–2008) confirms the existence of an inverted U-shaped relationship between government size and economic output.
- No robust linear relationship is found between government size and GDP growth; strong evidence supports a quadratic specification instead.
- The estimated optimal size of government in France is approximately 30% of GDP, reached in the late 1940s.
- This optimal share is higher than the 20% often found for the United States, but consistent with findings for other European countries.
- Results support the hypothesis that optimal government size varies across countries, implying country-specific inverted U-shaped curves depending on institutional design.

6 Paper 6: Economic growth and government size in developed European countries

6.1 Summary

Based on the literature on economic growth, there is a non-linear relationship between government size and economic growth, which is usually similar to an inverted U-shaped curve and is used to determine the optimum share of government expenditure. The present study aims to investigate the non-linear relationship among 14 developed European countries during 1995–2014. Final consumption expenditure to gross domestic product (FCE), current expenditure other than final consumption to gross domestic product (OCE), and government gross fixed capital formation to gross domestic product (GFCF) were considered for measuring government size. The results indicate an asymmetric effect of FCE and GFCF on economic growth when they are above and below the optimal level. The optimum values of FEC and GFCF were estimated to be 16.63 and 2.31However, it is noted that OCE always has a negative effect on economic growth. In terms of policy prescriptions, governments of developed countries should be aware that misallocation of public expenditure can occur given it may likely become unproductive after passing an optimal size. Hajamini and Falahi 2018

6.2 Metadata

1. **Link:** https://doi.org/10.1016/j.eap.2017.12.002
2. **Journal Name:** Economic Analysis and Policy
3. **Rank:** Q1

6.3 Methodology

$$
\begin{aligned}
\mathrm{GGDP}_{it} = \alpha_1 \mathrm{LAB}_{it} &+ \alpha_2 \mathrm{INV}_{it} + \alpha_3 \mathrm{EXP}_{it} + \alpha_4 \mathrm{IMP}_{it} \\
&+ (\delta_1 + \beta_1 \mathrm{GOV}_{it}) \cdot \mathbf{1}(\mathrm{GOV}_{it} \le \gamma_1) \\
&+ \sum_{j=2}^{J} (\delta_j + \beta_j \mathrm{GOV}_{it}) \cdot \mathbf{1}(\gamma_{j-1} < \mathrm{GOV}_{it} \le \gamma_j) \\
&+ (\delta_{J+1} + \beta_{J+1} \mathrm{GOV}_{it}) \cdot \mathbf{1}(\mathrm{GOV}_{it} > \gamma_J) \\
&+ u_{it}
\end{aligned}
$$

- GGDP_{it}: Annual growth rate of real Gross Domestic Product for country i at time t.
- LAB_{it}: Growth rate of working-age population for country i at time t.
- INV_{it}: Gross fixed investment as a proportion of GDP (proxy for the saving rate).
- INVP_{it}: Private fixed investment as a proportion of GDP (used when public investment is separated).
- EXP_{it}: Growth rate of exports of goods and services.
- IMP_{it}: Growth rate of imports of goods and services.
- GOV_{it}: Government size, measured as the share of government expenditure in GDP.
- FCE_{it}: Final consumption expenditure of government as a proportion of GDP.
- OCE_{it}: Other current government expenditure (excluding final consumption) as a proportion of GDP.
- GFCF_{it}: Gross fixed capital formation of the government as a proportion of GDP.
- α_0: Intercept term.
- $\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5$: Slope coefficients associated with explanatory variables.
- δ_j: Regime-specific intercept adjustment in the threshold model.
- β_j: Regime-specific slope coefficient of government size in the threshold model.

- γ_j: Threshold parameter determining regime changes.
- u_{it}: Composite error term.
- μ_i: Unobservable country-specific effect.
- λ_t: Unobservable time-specific effect.
- v_{it}: Idiosyncratic error term.
- $\mathbf{1}(\cdot)$: Indicator function that takes the value 1 if the condition holds and 0 otherwise.

6.4 Conclusions

- The study confirms the existence of the inverted U-shaped relationship between government expenditure and economic growth, known as the *Barro curve*, for 14 European Union countries.
- Unlike previous studies focusing only on Final Consumption Expenditure (FCE), this paper separately examines:
  - Final Consumption Expenditure (FCE)
  - Other Current Expenditure (OCE)
  - Gross Fixed Capital Formation (GFCF)
- Results show:
  - A non-linear (inverted U-shaped) relationship for FCE and GFCF.
  - A consistently negative effect of OCE on economic growth.
- The estimated optimal expenditure shares are:
  - FCE: approximately 16.6% of GDP.
  - GFCF: approximately 2.3% of GDP.
  - Combined optimal size (FCE + GFCF): approximately 19% of GDP.
- The optimal size of government expenditure appears to depend on income level, as results differ from previous findings for other countries.
- The confirmed non-linear relationship helps explain why prior empirical studies report conflicting positive and negative effects of government size on economic growth.

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